

PROJECT DESIGN PHASE-II

Data Flow Diagram & User Stories

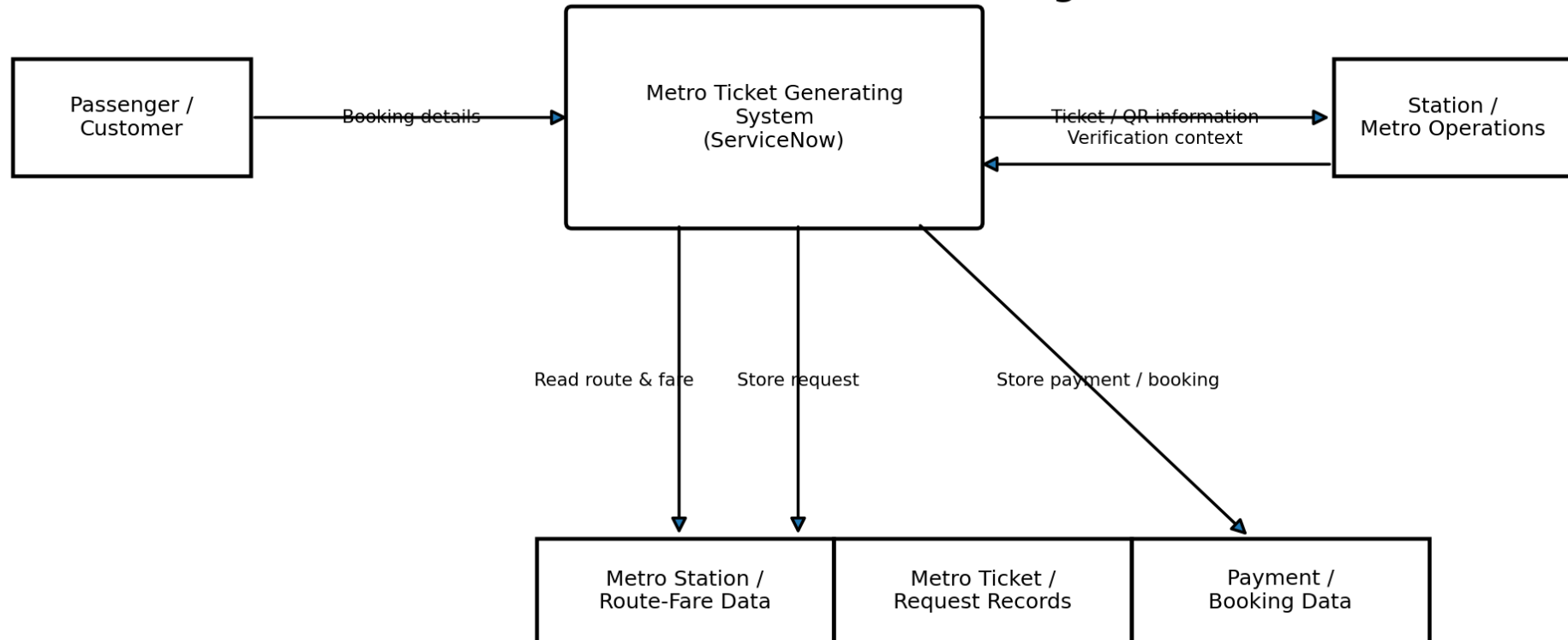
Metro Ticket Generating System

Date	Team ID	Project Name	Maximum Marks
04 September 2026	_____	Metro Ticket Generating System	4 Marks

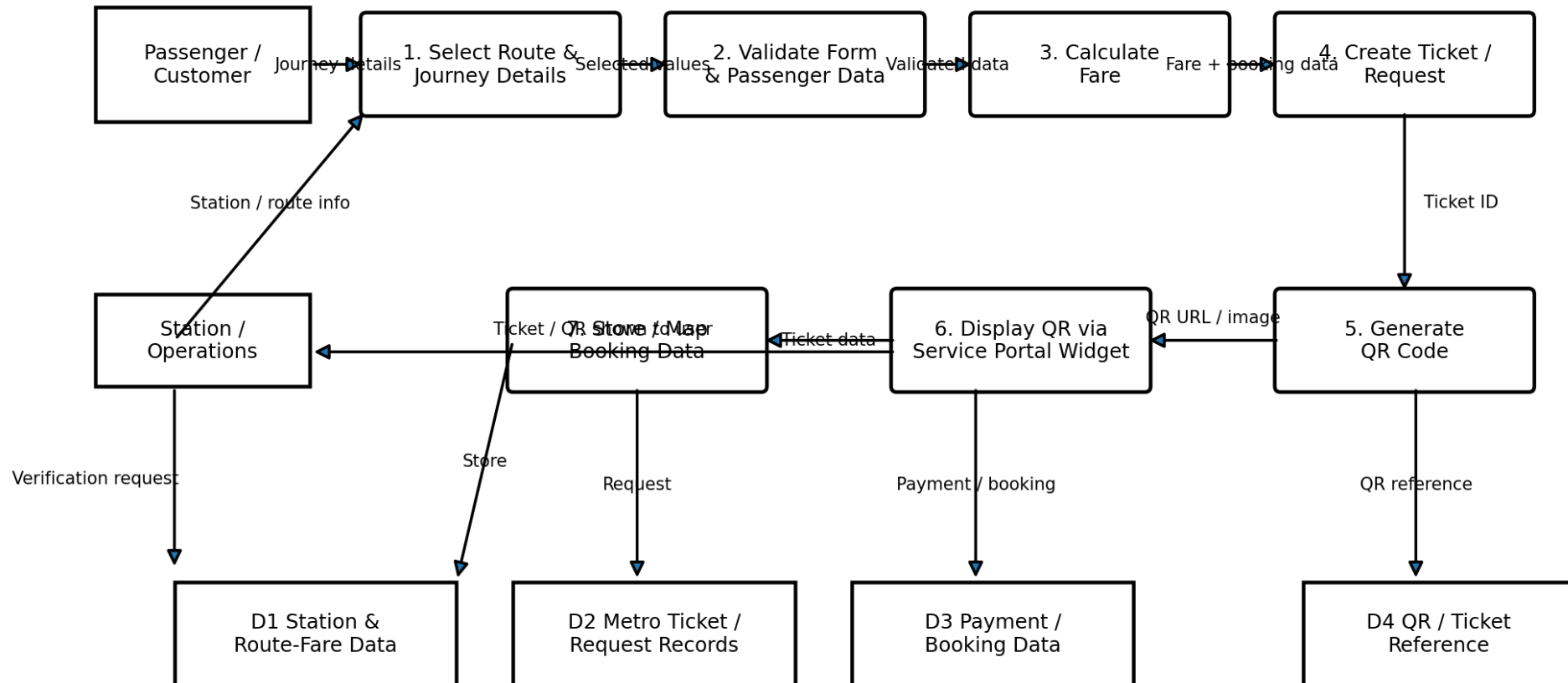
Data Flow Diagrams

The DFDs show how booking information enters the system, is validated and transformed into fare/ticket data, and how the QR ticket is presented to the passenger.

Level 0 - Context Data Flow Diagram



Level 1 - Metro Ticket Booking Data Flow



User Stories

The user stories follow the passenger booking flow and the administrator configuration/testing activities documented for the ServiceNow project.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Passenger / Customer	Ticket Booking	USN-1	As a passenger, I can open the Book a Metro Ticket catalog item and enter my journey details.	The booking form is accessible and displays the required journey fields.	High	Sprint-1
Passenger / Customer	Station Selection	USN-2	As a passenger, I can select the Starting From station.	A valid station can be selected from the reference list.	High	Sprint-1
Passenger / Customer	Station Selection	USN-3	As a passenger, I can select the Going To station.	A valid destination station can be selected from the reference list.	High	Sprint-1
Passenger / Customer	Journey Type	USN-4	As a passenger, I can choose Single Journey or Return Journey.	The selected journey type is retained and the applicable amount field is used.	High	Sprint-1
Passenger / Customer	Passenger Details	USN-5	As a passenger, I can select the number of passengers.	The passenger count is accepted from the configured choices and is used in fare calculation.	High	Sprint-1
Passenger / Customer	Fare Calculation	USN-6	As a passenger, I can see the fare calculated from my selected route, journey type and passenger count.	The appropriate fare is automatically populated; unrelated fare fields are cleared when applicable.	High	Sprint-2
Passenger / Customer	Payment	USN-7	As a passenger, I can select a payment mode.	Configured payment choices such as UPI, Card or Others are available.	Medium	Sprint-2
Passenger / Customer	Payment	USN-8	As a passenger, I can enter a payment mode detail when I select Others.	The Enter Payment Mode field becomes visible when Mode of Payment is Others.	Medium	Sprint-2
Passenger / Customer	Validation	USN-9	As a passenger, I can submit the request only after the required booking information is completed.	Mandatory and non-empty field validations prevent incomplete submissions.	High	Sprint-2
Passenger / Customer	QR Ticket	USN-10	As a passenger, I can receive a QR code after submitting my metro ticket request.	The submission flow generates a QR-code URL and presents the QR ticket to the user.	High	Sprint-3
Passenger / Customer	Portal Experience	USN-11	As a passenger, I can view my QR ticket through the Service Portal.	The Metro QR Widget displays the QR image and ticket message.	High	Sprint-3
Station / Metro Operations	Ticket Verification	USN-12	As a station/metro operations user, I can use the passenger's QR ticket as digital proof of the booking.	The passenger can present the generated QR ticket at the station for the intended verification process.	Medium	Sprint-3
Administrator	Catalog Administration	USN-13	As an administrator, I can configure the Book a Metro Ticket catalog item and its variables.	The catalog item contains the configured station, journey, passenger, fare and payment variables.	High	Sprint-1
Administrator	Automation	USN-14	As an administrator, I can configure Catalog Client Scripts and UI Policies for booking automation.	Fare automation, conditional field visibility and submission QR generation operate according to the configured logic.	High	Sprint-2
Administrator	Security	USN-15	As an administrator, I can protect backend ticket data using role-based access controls.	Customers can use the booking experience without direct access to restricted backend database records.	High	Sprint-3
Administrator	Testing & Quality	USN-16	As an administrator, I can test booking form behaviour, fare calculation, QR generation and visibility.	Configured test cases produce the expected form, fare and QR outcomes.	High	Sprint-4